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SolarSense AI - Detailed Concept Note

Project Title

SolarSense AI: Intelligent Rooftop Solar Recommendation and Assessment Platform

Background and Problem Statement

Solar adoption is often delayed because customers do not know the suitable system size, expected savings, installation cost, or return on investment. Traditional consultation requires site visits and manual calculations. SolarSense AI provides instant AI-powered preliminary solar assessments using only a few user inputs.

Vision

To democratize access to rooftop solar assessments using artificial intelligence and workflow automation.

Objectives

Provide instant recommendations, estimate system size, panel count, generation, savings, payback period, and improve awareness regarding renewable energy adoption.

Target Users

Residential homeowners, housing societies, small businesses, solar installation companies, consultants and government awareness programs.

Proposed Solution

Users enter name, email, pincode, roof dimensions and monthly electricity bill. The platform generates personalized solar recommendations and financial estimates.

System Architecture

Frontend: HTML/CSS/JavaScript on Netlify. Backend: n8n on Railway with persistent storage. AI: Google Gemini API through AI Agent. Data Flow: Frontend -> Railway Webhook -> n8n -> Gemini -> Frontend.

Technology Stack

HTML, CSS, JavaScript, n8n, Gemini API, Railway, Netlify, Docker, Git and GitHub.

Innovation

Combines AI and workflow automation, cloud-native deployment, persistent storage and low operational cost.

Implementation Journey

Initial deployment on Render failed due to database and memory issues. The project migrated to Railway using Docker and persistent volumes. Gemini integration was completed using AI Agent nodes.

Functional Requirements

User form, roof area calculation, webhook communication, AI recommendations and cloud deployment.

Non Functional Requirements

Availability, scalability, maintainability, reliability and security.

Expected Outputs

System capacity recommendation, panel count, monthly generation, savings estimate, payback period and personalized recommendation.

Future Enhancements

PDF reports, email reports, subsidy estimation, analytics dashboard, CRM integration and lead management.

Checklist

Completed: Netlify deployment, Railway deployment, persistent storage, public webhook, frontend-backend communication and Gemini integration. Planned: frontend rendering, email reports, analytics and PDFs.

Expected Impact

Reduce barriers to rooftop solar adoption and improve awareness regarding renewable energy investments.

Conclusion

SolarSense AI demonstrates practical use of AI, workflow automation and cloud infrastructure for renewable energy advisory systems.

Screenshots

 **SolarSense AI**

Agentic Solar Installation Advisor

Get Your Personalized Solar Report

Full Name

kds1

Email Address

ds@gmail.com

PIN Code

121212

Roof Dimensions

Length

655

Breadth

44

Unit

Feet

Calculated Roof Area: 28820.00 sq.ft.

Average Monthly Electricity Bill (₹)

40000

Generate Solar Report

Agentic AI N8N model

